

(N) is folklore-implicit: the Di Francesco–Kedem dictionary

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2026-10-02

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WIP commit: e877ce9 (this note plus the registry annotations it describes; the hash line was added in a follow-up commit).

Summary. Our Theorem (N) says that Hikita’s $\star$ is the $\mathcal{N}$ -transport of the ordinary product:

$$E_k = t^{-\binom{k}{2}} \mathcal{N} e_k \mathcal{N}^{-1}, \quad \mathcal{N} P_\nu(x; s, 1/t) = t^{n(\nu)} s^{n(\nu')} P_\nu(x; s, 1/t).$$

For all k it follows in about three lines from results *stated* in Di Francesco–Kedem, arXiv:1704.00154 (*CMP* 369 (2019) 867–928), hereafter [DFK]. [DFK] never state it for $k \geq 2$, so our verdict is **folklore-implicit**. This note gives the dictionary, the derivation with locators, the caveats, what remains ours, and a proposed reframing of the FPSAC story.

1. Dictionary

[DFK] use $\Gamma_i f = f(\dots, qx_i, \dots)$ and generalized Macdonald operators (their (1.5))

$$M_{\alpha; n} = \sum_{|I|=\alpha} x_I^n \prod_{i \in I, j \notin I} \frac{tx_i - x_j}{x_i - x_j} \Gamma_I.$$

They set $\eta = \exp(-\sum_i \text{Log}(Y_i)^2/2 \text{Log } q)$ and $\nabla^{(N)} := \eta^{-1}$ (Lemma 2.13, Rem. 2.14, (4.14)), with $\nabla^{(N)} P_\lambda = C_N t^{(N-1)|\lambda|/2 - n(\lambda)} q^{|\lambda|/2 + n(\lambda')} P_\lambda$. Under

$$(q, t)_{\text{DFK}} = (s, 1/t) :$$

- $P_\lambda^{\text{DFK}} = P_\lambda(x; s, 1/t)$ and $\Gamma_i = T_{s, i}$;
- $t_{\text{DFK}}^{-n(\lambda)} q^{n(\lambda')} = t^{n(\lambda)} s^{n(\lambda')}$, so $\mathcal{N}$ is the pure-eigenvalue part of $\nabla^{(N)}$; the extra factors C_N and $(t_{\text{DFK}}^{(N-1)/2} q^{1/2})^{\deg}$ are a scalar and a grading;
- $t_{\text{DFK}} x_i - x_j = t^{-1}(x_i - tx_j)$, hence $E_k = t^{k(N-k)} M_{k;1} \big|_{t_{\text{DFK}}=1/t}$.

2. The three-line derivation

1. By (6.5) and (6.8), $u_{0,j} = \frac{q^{j/2}}{1-q^j} p_j(X)$ and $u_{j,0} = \frac{q^{j/2}}{1-q^j} p_j(Y)$ carry the same normalisation. So $e_k(X) = \Phi_k(u_{0,\bullet})$ and $e_k(Y) = \Phi_k(u_{\bullet,0})$ with the same polynomial Φ_k .
2. The $SL_2(\mathbb{Z})$ action (6.3) has $U u_{0,j} = u_{j,j} = T u_{j,0}$, with $U \mapsto \tau_-$ and $T \mapsto \tau_+$. Hence $\tau_-(e_k(X)) = \Phi_k(u_{\bullet,\bullet}) = \tau_+(e_k(Y))$.
3. By Lemma 2.13 and Rem. 2.14, $\tau_- = \text{Ad } \nabla^{(N)}$. By Lemma 2.16 and Thm 2.17, $\tau_+(e_k(Y))|_{\text{Sym}} \propto M_{k;1}$. Therefore $\nabla^{(N)} e_k \nabla^{(N)-1} \propto M_{k;1}$; translated through §1, this is (N).

Numerical check (`scripts/day217dfk/dfk_check.py`). We took $N = 3$, exact arithmetic, $(q, t)_{\text{DFK}} = (3/7, -5/2)$, and every P_λ with $|\lambda| + k \leq 4$. With the pure eigenvalue $t^{-n} q^{n'}$, we found $\nabla e_k \nabla^{-1} = t_{\text{DFK}}^{-k(N-k) - \binom{k}{2}} M_{k;1}$ for $k = 1, 2, 3$; this is exactly (N) after the dictionary. With the full η^{-1} we found $\eta^{-1} e_k(X) \eta = \gamma^{-1} e_k(Y) \gamma$ on Sym_N , with no scalar. Negative control: the opposite conjugation $\nabla^{-1} e_k \nabla$ is *not* proportional to $M_{k;1}$.

3. Caveats

- [DFK] never write step 2 or the conclusion for $k \geq 2$. In §6.3 they conjugate only by T , never by U . Their $k = 1$ statement, Rem. 6.2 ($\nabla^{(N)} u_{a,b} \nabla^{(N)-1} = u_{a+b,b}$), rests on the Schiffmann–Vasserot SL_2 -equivariance of the EHA $\rightarrow$ spherical DAHA map. They quote this from SV11/Che05 rather than prove it.
- The literal proof of their Lemma 2.13 (apply the anti-involution ε to Lemma 2.12) yields $\text{Ad } \eta$, not $\text{Ad } \eta^{-1}$: $\varepsilon(\gamma^{-1} \varepsilon(b) \gamma) = \eta b \eta^{-1}$. Our numerics confirm that the *stated* direction η^{-1} is the one that produces $M_{k;1}$.
- The case $k = 1$ is also in BGHT 1999, (I.12)(iii): $\nabla e_1 \nabla^{-1} = -D_1$ (plethystic, infinitely many variables). For general k , BGLX arXiv:1405.0316, Conj. 2.1, identifies $\nabla e_k \nabla^{-1}$ with Negut’s $N_{k,k}$ (plethystic). We have not checked its later status.
- Hikita 2503.23597 never mentions ∇ . His Def. 3.4 already *defines* $\star$ as transport by $\mathbf{q}(F) = F(Y) \bullet 1$, so the content of (N) is “ $\mathbf{q} = \nabla^{(N)}$ up to normalisation”.

4. What remains ours

- (i) The **elementary, self-contained proof**. It uses Lemma (I), $[e_1(Y), X_1] = (s-1)X_1Y_1$, and the nonsymmetric Gaussian $\hat{\gamma}X_i\hat{\gamma}^{-1} = Y_i^\bullet$. It avoids EHA/SV11 entirely. [DFK] contain no nonsymmetric statement of this kind and nothing like Lemma (I).
- (ii) The **identification with Hikita’s $\star$** . Neither side states it.
- (iii) The **edge theorems**:
 - H: as $s \rightarrow 0$, $\star$ becomes Hall–Littlewood e_k -Pieri in the rescaled basis b_μ ;
 - H’: as $t \rightarrow \infty$, the limit is q-Whittaker, $P_{\mu'}(x; s, 0) = \omega Q'_\mu$;
 - the $t = 1/s$ content-twisted Littlewood–Richardson line. It is now proved as a corollary of (N) and checked computationally in 41/41 cases (`scripts/day217/fast.py`). Its novelty is also weak.

H and H’ are 3-line corollaries of (N), so they are now also derivable from [DFK]. Their novelty rests on the explicit identification of the limits, not on the transport.

The registry is annotated accordingly (`registry/hikita-star-dominance-support.json`, field `novelty_2026_10_02` on nodes `N-star-is-nabla-transport`, `theorem-H-s0-star-is-HL-pieri`, `theorem-H-prime-t-infinity-q-whittaker`). Trust grades are unchanged.

5. Proposed reframing of the FPSAC story

We propose to stop presenting (N) as the headline.

- **Lead with the explicit edges**: H (HL), H’ (q-Whittaker), (KF), and dominance support (DS) with its exact up-set and the zeta-function limit at $s = t = 0$.
- **Cite [DFK] for the transport**. State (N) as “Hikita’s level-one twist is Cherednik’s τ_+ , so $\star$ is the $\nabla^{(N)}$ -transport of the ordinary product [DFK, Thm 2.17, L2.12–2.13, Rem. 2.14, (4.14)]; cf. BGHT (I.12)(iii)”.
- **Offer our proof as an elementary alternative**: Lemma (I) and the nonsymmetric Gaussian, which avoid the EHA.

The story becomes “explicit degenerations of a ∇ -twisted product”.

Request. Please check the dictionary in §1, in particular $(q, t)_{\text{DFK}} = (s, 1/t)$ and $E_k = t^{k(N-k)} M_{k;1}$, and the η vs. η^{-1} wrinkle in §3. You can skip a from-scratch novelty hunt on $\mathcal{N}$.